MF-BHNet: A Hybrid Multimodal Fusion Network for Building Height Estimation Using Sentinel-1 and Sentinel-2 Imagery

Siyuan Wang, Bowen Cai, Dongyang Hou, Qing Ding, Jiaming Wang, and Zhenfeng Shao

***Abstract*— Integrated Sentinel-1 synthetic aperture radar (SAR) imagery and Sentinel-2 optical imagery have shown great promise in mapping large-scale building height. Effectively fusing the complementary features of SAR and optical imagery is a key challenge in enhancing the building height estimation performance. However, SAR imagery and optical imagery have significant heterogeneity, which makes obtaining accurate building height a challenging problem. In this article, we propose a hybrid multimodal fusion network (MF-BHNet) for building height estimation using Sentinel-1 SAR imagery and Sentinel-2 optical imagery. First, we design a hybrid multimodal encoder to mine modal-specific feature and model intermodal correlation. In particular, an intramodal encoder (IME) is designed to reconstruct valuable intramodal information, and a transformer-based cross-modal encoder (CME) is used to model intermodal correlation and capture contextual information. Then, a coarse-fine progressive multimodal fusion method is proposed to fuse SAR feature and optical feature to improve the building height estimation performance. We construct a building height dataset by introducing superior building footprints to validate our method. Experimental results demonstrate that our MF-BHNet method outperforms the compared 11 state-of-the-art methods, which achieves the lowest root-mean-square error (RMSE) of 3.6421 m. Besides, compared to the four publicly available building height products, the mapping result of the proposed method has significant advantages in terms of spatial detail and accuracy.**

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I. Introduction

BUILDING height characterizes urban development in vertical structure, which influences urban volume ratio, material stock, and built-up environment [1], [2]. It is also the key to evaluating the urban ecological environment and development status and plays an important role in analyzing urban landscape structure [3], evaluating urban human livability [4], and estimating economic development patterns [5]. Especially in recent years, with the accelerated urbanization process in developing countries, the increase of regional high-rise and high-density buildings has also caused some socio-environmental problems, such as increased energy consumption, urban congestion, and residential segregation [6], [7], [8]. Therefore, accurate mapping of building height over large areas is essential for a comprehensive understanding of urban development processes.

With the development of Earth observation technology, increasing attention has been paid to the extraction of building height information using remote sensing data. In many cases, building height can be measured via ultrahigh-resolution stereo image pairs [9], [10], but collecting large-scale stereo images is expensive. Furthermore, building height can be determined using the building shadow length in optical remote sensing imagery. [11], [12]. However, shadow extraction is susceptible to occlusion and atmosphere. Recently, the quantitative analysis of Sentinel satellite constellation has provided new perspectives on building height estimation, with the advantage of global coverage as well as the synergy of Sentienl-1 synthetic aperture radar (SAR) and Sentinel-2 optical imagery [5], [13], [14]. Machine learning methods, such as the support vector machine and random forest, are increasingly used for large-scale building height mapping with Sentinel-1/2 data [15], [16]. However, these methods are inadequate for mining the deep nonlinear features of multisource data. It is a challenge for performance largely depends on the quality of the feature engineering.

Deep learning, especially convolutional neural networks (CNNs), which learn the representation of images layer by layer, has significant advantages in urban information extraction and large-scale mapping [17], [18], [19], [20]. Inspired by this, several studies have explored the use of deep learning to collaborate with Sentinel-1/2 data for building

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height estimation [21], [22], [23]. Capturing effective SAR and optical fusion feature is the key to facilitating the building height estimation performance. However, current feature cascading [23] or band cascading [22] is loosely feature fusion, which not only results in information redundancy but also makes it difficult to eliminate the discrepancy between different modalities. Besides, valuable feature perception is the basis for effective fusion, and the current methods are insufficient to understand the critical information of a single modality.

To this end, this article proposes a hybrid multimodal fusion network (MF-BHNet) for building height estimation. First, a hybrid multimodal encoder is proposed for mining intramodal feature and modeling intermodal correlation. Specifically, we design an intramodal encoder (IME) by introducing optimal convolutional units to capture modality-specific critical information and reduce redundancy. Furthermore, a transformer-based cross-modal encoder (CME) is used to model intermodal correlation and perceptual contextual information. Then, a coarse-fine progressive multimodal fusion method is proposed to fuse the complementary information of SAR images and optical images to improve the precision of building height estimation. Thus, the semantic, spatial detail, and height response information of the building are fused at different levels, which not only helps in estimating the building height but also improves the quality of the building footprint.

The main contributions of this article can be summarized as follows.

1) We propose a novel hybrid multimodal fusion network for estimating building height combined Sentinel-1 SAR imagery and Sentinel-2 optical imagery. With the global coverage of the Sentinel constellation, the method has the potential to produce large-scale building height maps.
2) A hybrid multimodal encoder is designed to mine modal-specific feature and model intermodal correlation. In this way, the bottleneck of multimodal fusion due to the insufficient representation of unimodal information can be avoided.
3) A coarse-fine progressive multimodal fusion method is proposed to bridge the heterogeneity of SAR and optical imagery. By multiscale feature fusion (MFF) and multimodal semantic fusion (MSF), the discrepancy of heterogeneous data is effectively eliminated, promoting the exploitation of complementary information.
4) A building height dataset is constructed, and extensive comparative experiments are carried out with the other 11 state-of-the-art methods. Experimental results indicate that our method has a mean error of 3.6421 m, which is 27.6% better than the suboptimal method. In addition, comparing the four publicly available building height products, our mapping has a significant superiority in detail.

The rest of this article is organized as follows. Section II provides an overview of related work. Section III outlines the proposed methodological framework. Experimental setup is presented in Section IV. Section V describes our extensive experiments and analysis. Finally, Section VI summarizes the conclusion.

II. Related Work

In this section, we review work closely related to this article, including building height estimation using optical remote sensing imagery and large-scale building height mapping using SAR imagery from the Sentinel-1/2 constellation. The former mainly utilizes high-resolution optical images, which are estimated by monocular measurement or stereoscopic estimation. The latter tends to favor national/planetary scale mapping studies via Sentinel-1 SAR imagery and Sentinel-2 optical imagery.

A. Building Height Estimation Using Optical Remote Sensing Imagery

With the development of computer vision, deep learning is applied to building height estimation from optical remote sensing imagery. According to the estimation principle, we can divide them into monocular image estimation-based methods and multiview/stereo image-based methods.

1) Monocular Image Estimation-Based Methods: The idea of using monocular images for building height estimation comes from monocular depth estimation in computer vision, which aims to estimate the depth of objects in a scene from a single 2-D image. In fact, both involve recovering depth information from 2-D image projections of a 3-D scene. Inspired by this, several studies have explored the application of CNN to superhigh-resolution optical images [24], [25] and street view images [26]. For example, Liu et al. [25] constructed a regression model with an encoder–decoder structure to learn the mapping relationship from a single aerial image to the digital surface model (DSM) using the 0.15-m spatial resolution optical images. Ureña-Pliego et al. [26] attempted to learn the building height information from a single streetscape image based on the segmentation model. These methods do not require additional auxiliary data and learn the implicit mapping relationship between 2-D and 3-D. However, it is still a very challenging task due to the poor interpretability of the mapping relationships and the limited generalization ability of the model.

2) Multiview/Stereo Image-Based Methods: It mainly exploits the inherent relationship between multiview/stereo images and building height to drive the optimization of deep learning models. For example, Cao and Huang [27] proposed a multispectral, multiview, and multitask deep network for building height estimation, introducing a 2.5-m spatial resolution ZY-3 multiview imagery to reduce the uncertainty of building height estimation. Using the stereo observation capability of China's GaoFen-7 satellite, Chen et al. [28] first generate DSM data from rear-view and front-view images. Then, building roofs are combined with a semantic segmentation network to generate building roofs, and finally, the height of each building is estimated based on the DSM data and building outline data. Relying on superhigh-resolution satellite imagery, these

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methods are able to map fine building height with exciting results. However, it is difficult to acquire superhigh-resolution multiview/stereo imagery, which is currently not applicable to large-scale mapping tasks. Therefore, the above studies have only been validated in a few cities.

B. Large-Scale Building Height Mapping Using Sentinel-1 SAR Imagery and Sentinel-2 Optical Imagery

The Sentinel-1/2 satellites provide publicly available 10-m spatial resolution imagery over global coverage. It achieves an ideal balance between spatial coverage and resolution for large-scale mapping needs [29], [30]. Moreover, the backscatter values of Sentinel-1 SAR data correlate with building height [31]. Inspired by this, Li et al. [5] constructed a building height indicator called VVH using the polarization bands VV and VH of Sentinel-1 SAR imagery. Yang and Zhao [16] developed a building height dataset with a spatial resolution of 1 km for China in 2017, using a spatially informed Gaussian process regression model. However, SAR imagery lacks clear spatial details, which makes it difficult to perform fine-scale building height mapping using SAR data by itself.

To capture detailed building boundaries, several studies have explored using complementary information from SAR and optical imagery to map the building height. For example, Frantz et al. [32] applied a support vector model to map the building height in Germany by integrating the shading features of Sentinel-2 optical imagery and the backscatter features of Sentinel-1 SAR imagery. Li et al. [15] combined Sentinel-1 SAR data, Landsat optical imagery, and other ancillary data to map 3-D building structures based on a random forest model. Considering that deep learning has powerful feature-fitting capabilities, some studies have attempted to use deep learning to collaborate with Sentinel-1/2 data for joint mapping [21], [22], [23]. Yadav et al. [21] used two CNNs to capture the features of SAR and optical images separately, followed by cascading various levels of features to decode building height maps. Li et al. [22] developed a multitask network for building height mapping at a 100-m resolution, utilizing all the bands of SAR and optical images as combined inputs. Further, Cai et al. [23] proposed a building height estimation network (BHE-Net) for 10-m resolution mapping, which cascades the features of the two modalities at a high level, followed by refining the features using an attention mechanism.

In summary, the Sentinel data facilitate building height mapping over large areas. Band cascading or feature cascading is commonly used in Sentinel-1/2 imagery for collaborative building height mapping. However, SAR imagery and optical imagery have significant heterogeneity, and previous coarser methods do not fully exploit the multimodal information. Moreover, the current methods are insufficient to extract critical information from a single modality, which is a major limitation to the effectiveness of fusion. Therefore, how to design a reasonable fusion strategy is a key issue to be solved. It is necessary to fully exploit the complementary information of multimodal data while taking into account the modality-specific information.

III. Methodology

Given a paired Sentinel-1 SAR image x_1 and Sentinel-2 optical image x_2, with the same spatial extent, where $x_1 \in R^{2\times H\times W}$, the spatial resolution is $H \times W$, and the number of channels is 2 representing the two polarization bands VV and VH. The optical image $x_2 \in R^{4\times H\times W}$, which includes red, green, blue, and near-infrared (NIR) bands. Our goal is to predict the corresponding pixel-level building height map of size $H \times W$.

A. Overview

The overall framework of the proposed MF-BHNet is shown in Fig. 1, which is a dual-branch multitask learning network. MF-BHNet takes SAR imagery and optical imagery as inputs to each branch and gets the height and footprint of the building through the two prediction branches. In the workflow, we first use an IME to obtain the optimized modality-specific critical features from SAR and optical images, respectively. Internally, a multimodal feature-semantic fusion module is designed to fuse SAR and optical features from coarse to fine, combining MFF and MSF. Then, a transformer-based CME is proposed to model intermodal associations via global context learning. Specifically, MF-BHNet consists of four components: IME, CME, multimodal feature-semantic fusion module, and decoder.

The IME is designed as a dual-branch CNN to extract critical modality-specific details from SAR and optical data, respectively. In each branch, we design a new residual unit to reduce information redundancy. We adopt unshared weights to learn domain-specific information for a single modality.

The CME is implemented as a transformer for globally modeling intermodal correlation. It encodes unimodal feature maps and semantic fusion modal feature maps into patches to produce continuous inputs. By modeling the global context on the top of the CNN features, the problem of losing spatial details in the pure transformer is avoided.

The multimodal feature-semantic fusion module facilitates the transfer and interaction of multimodal information at both the feature and semantic levels. In particular, the MFF utilizes a feature cascade to convey the low-level spatial structure information of the building. The MSF first refines the semantic features of a single modality and then performs channel shuffling of multimodal features to achieve semantic fusion.

The decoder is designed to convert multiscale features into building height and building footprints. It follows the same twinned design with unshared weights as the local feature encoder. The decoder up-samples global context-encoded features and combines them with high-resolution local feature maps. In this way, low-level spatial structural features are fully exploited with high-level semantic information.

In general, MF-BHNet attends to both intramodal critical information extraction and intermodal hierarchical level fusion. Further experiments show that our design can effectively facilitate the exploitation of SAR images and optical images. The following describes the hybrid multimodal encoder, the multimodal feature-semantic fusion module, the decoder, and the loss function in detail.

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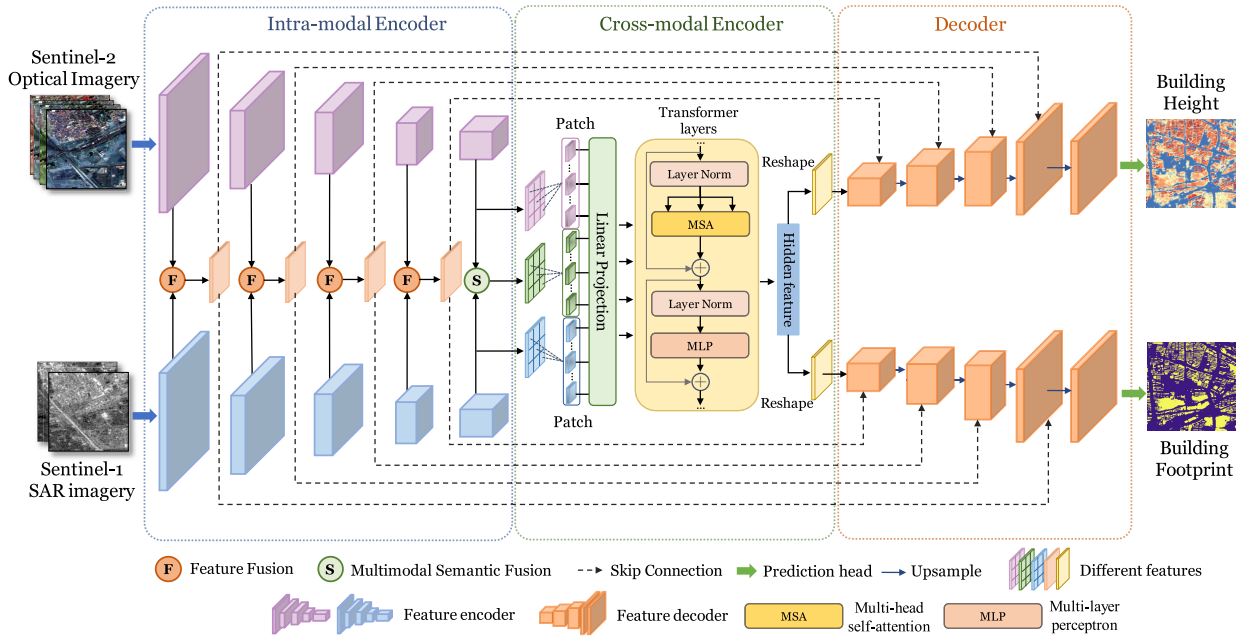


Fig. 1. Overall framework of the MF-BHNet. The IME aims to capture single modal key features, and the CME uses transformer to model intermodal correlation, and the decoder is utilized to generate predictions. Inside the network, multiscale feature fusion and MSF modules are learned together with the main structure.

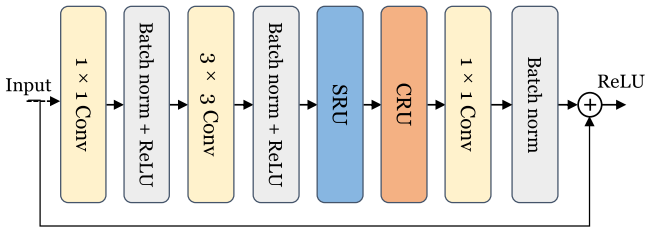


Fig. 2. SC-RU diagram. SRU and CRU are sequentially embedded within the residual block.

B. Hybrid Multimodal Encoder

1) Intramodal Encoder: In previous studies, the twinned ResNet network [33] is generally constructed as an encoder to capture the multilevel local feature representations of different modalities. However, due to the 10-m spatial resolution of the sentinel image, a large number of mixed image pixels are present. The spectral signals of objects may mix with each other, which may cause the network to learn some incorrect target distributions and redundant information.

To address this issue, we design a residual unit SC-residual unit (SC-RU) for constructing an IME, which reconstructs critical feature and reduces redundant information. As illustrated in Fig. 2, the SC-RU incorporates the state-of-the-art spatial reconstruction units (SRUs) and channel reconstruction units (CRUs) [34] into the residual block to refine information in both spatial and channel dimensions. Considering that rich information in remote sensing images tends to be at lower levels, such as texture attributes and elevation responses. Therefore, we embed the SRU and CRU within the residual blocks to handle more diverse information.

Fig. 3 shows the structure of the SRU. The SRU utilizes separation and reconstruction to generate spatial reconstructed feature maps. Initially, the information content of the feature map is evaluated using the scale factor in the group normalization (GN) layer. Then, the weight values of the feature maps are gated by a threshold to obtain the information-rich feature map and the information-poor feature map. Finally, the information-rich features are combined with the information-poor ones using a cross-reconfiguration operation to generate more informative features.

The CRU process is shown in Fig. 4. It implements a separation–transformation–fusion strategy to reduce channel redundancy. First, the CRU splits the spatial refinement features from the SRU into two feature maps with different numbers of channels, i.e., αC and $(1-\alpha)C$. Afterward, the CRU applies different convolutions to transform them and obtain features with rich and scarce channel information. Finally, pooling and elementwise operations are sequentially used to merge the information from different channels to obtain the channel-refined feature.

With efficient convolutional reconstruction, SC-RU not only can improve the discriminative ability of heterogeneous features but also further reduce the computational complexity. Further, by stacking the SC-RU residual blocks, the IME can achieve a better representation capability. In summary, the feature extraction process of local feature encoder can be described as

$$\begin{cases} F^s = E^s(x_1) = E_3^s\left(E_2^s\left(E_1^s\left(C_{7\times7}^s(x_1)\right)\right)\right) \\ F^o = E^o(x_2) = E_3^o\left(E_2^o\left(E_1^o\left(C_{7\times7}^o(x_2)\right)\right)\right) \end{cases} \tag{1}$$

where E^s denotes the SAR image encoder, E^o denotes the optical image encoder, and E_i^s and E_i^o denote the corresponding residual block, respectively, $i \in \{1, 2, 3\}$. In addition, $C_{7\times7}^s$ and $C_{7\times7}^o$ denote the 7×7 convolution.

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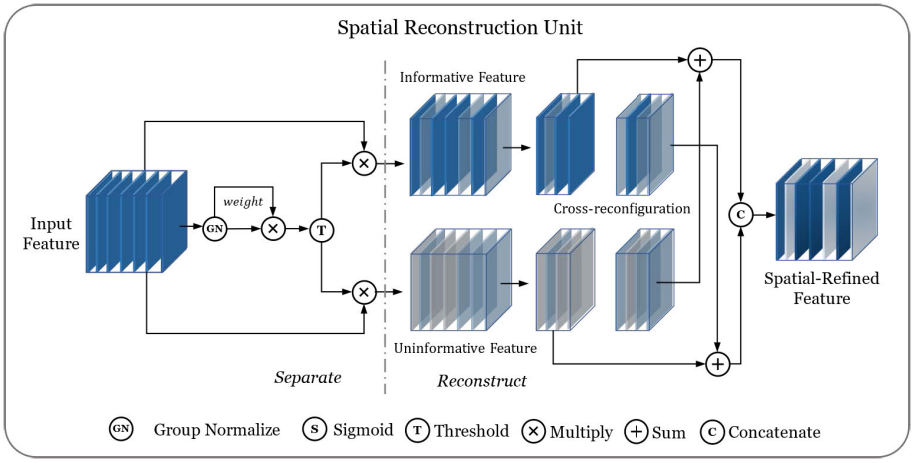


Fig. 3. Structure of SRU.

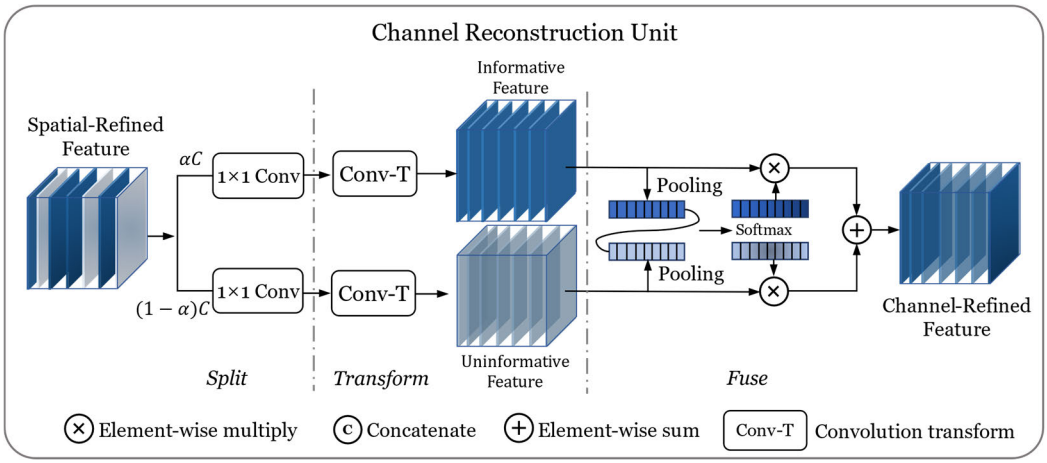


Fig. 4. Structure of CRU.

To obtain multiple feature maps with four different resolutions, the output of the 7 × 7 convolution and various residual blocks are retained. These feature maps are denoted as $\{F_1^o, F_2^o, F_3^o, F_4^o\}$ and $\{F_1^s, F_2^s, F_3^s, F_4^s\}$ from bottom to top. In particular, $F_1^s \in R^{(H/2)\times(W/2)\times D_1}$, $F_2^s \in R^{(H/4)\times(W/4)\times D_2}$, $F_3^s \in R^{(H/8)\times(W/8)\times D_3}$, and $F_4^s \in R^{(H/16)\times(W/16)\times D_4}$, where D_1, D_2, D_3, and D_4 represent the number of channels, respectively. Note that the SAR- and optical-derived feature maps have the same shape.

2) Cross-Modal Encoder: Beyond intramodal learning, we design a CME to model the intermodal correlation and facilitate the interaction of multimodal information. The CME is driven from a transformer [35], which can process the multimodal input sequences and generates the output sequences by self-attention. By globally capturing features from multiple modalities simultaneously, CME can learn remote dependencies of multimodal data. As shown in Fig. 1, CME requires three different feature maps provided by the IME, which are the optical modal feature map, SAR modal feature map, and MSF feature map, defined as F_o, F_s, and F_m, respectively.

Initially, CME serializes these feature maps into different fixed 2-D patches $\{o_p^i \in R^{P^2\cdot C} | i = 1, \ldots, n\}$, $\{s_p^i \in R^{P^2\cdot C} | i = 1, \ldots, n\}$, and $\{m_p^i \in R^{P^2\cdot C} | i = 1, \ldots, n\}$, where C denotes the channel count, P denotes the patch size, and $n = (HW/P^2)$ denotes the number of patches. We aggregate these patches together to obtain a multimodal aggregation representation, denoted as $\{f_p^i \in R^{P^2\cdot C} | i = 1, \ldots, N\}$, where N is the total number of all patches.

Afterward, the learnable positional embedding information is assigned to each block as input. The vectorized patch f_p^i is mapped to a potential D-dimensional embedding space through a linear projection. To incorporate spatial information, the patch embedding is augmented with position information, which is defined as

$$g_0 = \left[f_p^1 e; f_p^2 e; \ldots; f_p^N e\right] + e_{\text{pos}} \tag{2}$$

where $e \in R^{(P^2\cdot C)\times D}$ denotes the block embedding projection, and $e_{\text{pos}} \in R^{N\times D}$ denotes the positional embedding.

Finally, we perform self-attention encoding. The transformer consists of L self-attention blocks that capture contextual dependencies. Each self-attention block incorporates a multihead self-attention (MSA) and a multilayer perceptron (MLP). The output of layer l can be expressed as

$$g_l' = \text{MSA}(\text{LN}(g_{l-1})) + g_{l-1} \tag{3}$$

$$g_l = \text{MLP}\left(\text{LN}\left(g_l'\right)\right) + g_l' \tag{4}$$

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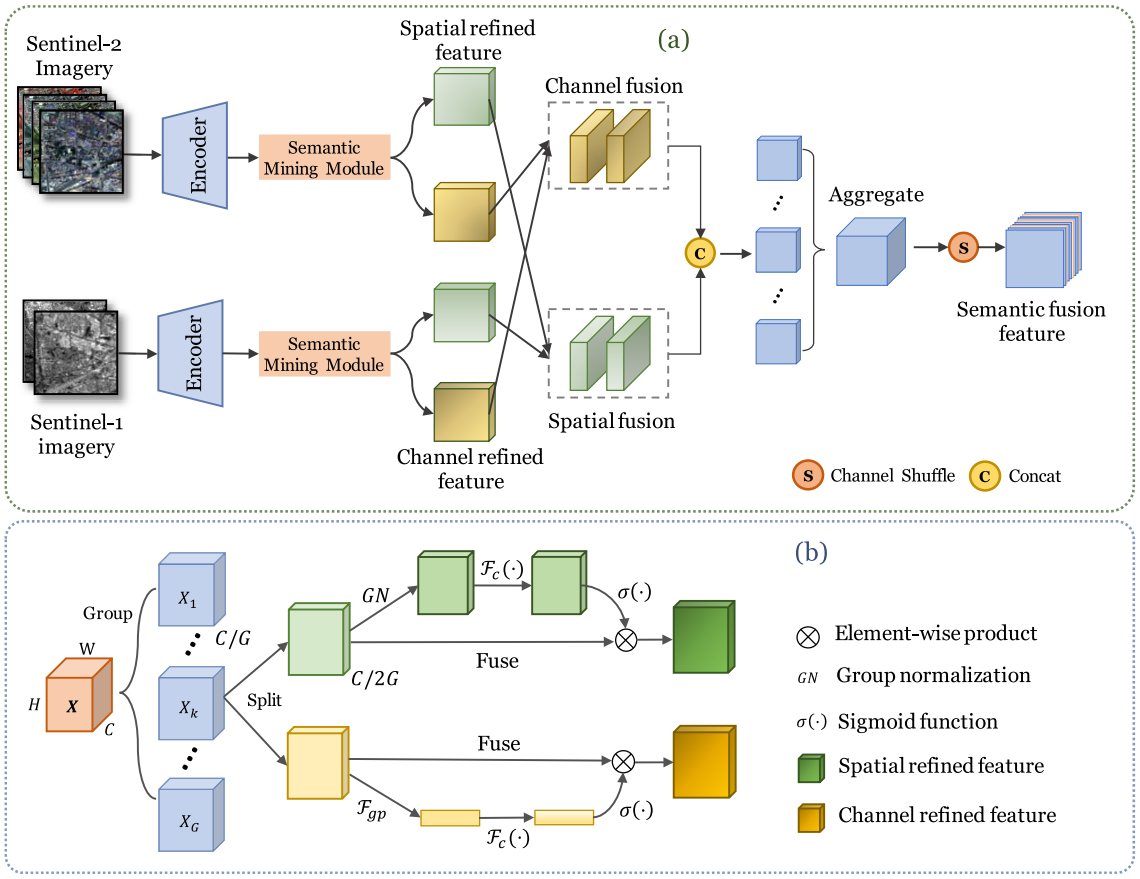


Fig. 5. Framework of MSF, where encoder represents the IME. (a) MSF module. (b) SSM.

where LN(·) is the layer normalization, and g_l is the coded image representation. In this way, the output of the last self-attention block is regarded as a global representation with the shape $(HW/P^2) \times D$. We reshape it to recover the spatial order, denoted as $G \in R^{(H/P)\times(W/P)\times D}$.

It should be noted that the transformer is powerful in global context modeling, but it ignores spatial details at lower levels of the image. In contrast, we solve this problem by embedding high-level feature maps from CNN into patches, which is able to learn both local details and global contextual information. With the cross-modal global sensing of CME, the intermodal correlation between SAR data and optical data is effectively captured.

C. Multimodal Feature-Semantic Fusion

Optical imagery provides spectral and textural information, while SAR imagery contains valuable geometric information about ground objects, which are highly complementary. Joint optical and SAR imagery can help detect the contours of buildings. More importantly, the combination of the two sensors improves the ability of dual-polarized radar backscatter to estimate the building height [32]. However, the current building height estimation methods rarely consider fine-grained multimodal fusion. To solve this problem, we propose a multimodal progressive feature-semantic fusion method to improve the performance of SAR images and optical images in building height estimation. In the following, we present MFF and MSF, respectively.

1) Multiscale Feature Fusion: MF-BHNet uses CNN as a local feature encoder, which represents image features at multiscale levels, and low-level feature maps contain rich detail information [36], such as texture, edge, and height. In general, low-level features are usually relatively coarse; thus, access to a wider range of information is often more important. Therefore, we adopt a MFF strategy that uses feature cascades to increase the information of the fused feature map with lower complexity.

We first concatenate the feature maps of the SAR branch and the optical branch and then use a 1×1 convolution to reduce the channel dimensions to obtain a preliminary feature map. Afterward, batch normalization and ReLU activation are performed to improve the representation. The four different levels of features obtained by the local feature encoder are $\{F_1^o, F_2^o, F_3^o, F_4^o\}$ and $\{F_1^s, F_2^s, F_3^s, F_4^s\}$. In particular, the ith level features from optical and SAR images are denoted as F_i^o and F_i^s, $i \in [1, 2, 3, 4]$, respectively. The feature fusion can be built as

$$F_i' = \mathrm{ReLU}\left(\mathrm{BatchNorm}\left(C_{1\times1}\left(F_i^o \odot F_i^s\right)\right)\right) \tag{5}$$

where F_i' denotes the fused features, $\odot$ denotes the feature splicing along the channel, and $C_{1\times1}$ denotes 1×1 convolution. We perform feature fusion on the four levels to obtain a multiscale fusion feature, denoted as $\{F_1', F_2', F_3', F_4'\}$.

It is noteworthy that in the previous description, we design the SC-RU structure to reduce redundancy, so that MFF does

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not cause a large amount of redundancy. It is a simple yet effective fusion strategy.

2) Multimodal Semantic Fusion: The aforementioned MFF is a relatively coarse approach aimed at obtaining richer multimodal information. However, to achieve a more detailed semantic description of buildings, it is necessary to perform a fine-grained fusion of information from different modalities. To this end, we design an MSF module that facilitates the extraction of intramodal discriminative information while taking into account fine-grained cross-modal interactions.

The MSF strategy is inspired by [37] and employs an attention mechanism to refine the modal-specific feature. The diagram of MSF is shown in Fig. 5(a). First, the optical feature and SAR feature are refined with a semantic mining module (SMM) to highlight valuable spatial and channel features and suppress redundant information. Subsequently, the spatial and channel fusion strategies are utilized to combine the spatial and channel features of different modalities, respectively. Finally, a channel shuffle operation is introduced to facilitate intermodal information interaction.

Specifically, an attention mechanism-driven SMM is introduced to capture the key modal-specific feature, as shown in Fig. 5(b). Two independent SMMs are used to model the spatial attention and channel attention of each modality, respectively. The processing flow of each SMM branch is described as follows.

To simplify the description, we define the input to the SMM as a feature map X with C channels and a 2-D shape of $H \times W$. First, X is divided into G groups at the channel dimension, i.e., $X = \{X_1, \ldots, X_k, \ldots, X_G\}$, where $X_k \in R^{C/G\times H\times W}$. Then, the sub-feature X_k of the grouping is separated into two streams in the channel dimension, i.e., $X_{k1}, X_{k2} \in R^{C/2G\times H\times W}$. One processing stream captures critical channel information, while the other stream acquires spatially focused regions. As for the channel attention stream, we first embed global information using global average pooling to generate the channel statistic $s \in R^{C/2G\times 1\times 1}$, which is computed as follows:

$$s = F_{\text{gap}}(X_{k1}) = \frac{1}{H\times W}\sum_{m=1}^{H}\sum_{n=1}^{W} X_{k1}(m,n). \tag{6}$$

Then, the channel refined feature is obtained through a simple gating mechanism $F_c(\cdot)$ with an s-shaped activation function. It can be formulated as

$$X'_{k1} = \sigma(F_c(s))\cdot X_{k1} = \sigma(W_1 s + b_1)\cdot X_{k1} \tag{7}$$

where X'_{k1} denotes the channel refinement result, $W_1 \in R^{C/2G\times 1\times 1}$ and $b_1 \in R^{C/2G\times 1\times 1}$ are the parameters used to scale and shift s to enhance the feature representation of X_{k1}, and $\sigma(\cdot)$ is the Sigmoid function. To refine the spatial feature, we first obtain spatial statistics by applying GN to X_{k2}. Then, the spatial refined output is computed as

$$X'_{k2} = \sigma(W_2\cdot \text{GN}(X_{k2}) + b_2)\cdot X_{k2} \tag{8}$$

where X'_{k2} denotes the spatial refined feature, and $W_2 \in R^{C/2G\times 1\times 1}$ and $b_2 \in R^{C/2G\times 1\times 1}$ are used for scaling and shifting to enhance the feature representation of X_{k2}.

By capturing the modality-specific critical feature through semantic mining, multimodal high-level information can be fused more efficiently. We define X^s_{k1} and X^o_{k1} to represent the channel-refined features of SAR imagery and optical imagery, respectively. Similarly, X^s_{k2} and X^o_{k2} represent the spatial-refined features of SAR imagery and optical imagery, respectively. The features of different modal are then fused together through channel fusion and spatial fusion. We cascade the channel feature X^s_{k1} and X^o_{k1} to obtain the channel fusion feature $X^m_{k1} = (X^s_{k1} \odot X^o_{k1}) \in R^{C/G\times H\times W}$, where $\odot$ represents the channelwise concatenation. Similarly, the spatial fusion feature $X^m_{k2} = (X^s_{k2} \odot X^o_{k2}) \in R^{C/G\times H\times W}$ is obtained.

To combine the spatial and channel features, we cascade X^m_{k1} and X^m_{k2} to obtain the sub-feature $X^m = (X^m_{k1} \odot X^m_{k2}) \in R^{2C/G\times H\times W}$. Afterward, the sub-feature is aggregated and channel shuffle is performed to obtain the semantic fusion feature $S \in R^{2C\times H\times W}$. In this way, cross-group information is allowed to flow along the channel dimension, which enables fully exchanging and fusing information from both modalities. Since deeper features contain more significant semantic information, we perform MSF between the highest SAR feature F^s_4 and the highest optical feature F^o_4. The semantic fusion feature S_4 can be computed as

$$S_4 = \text{MSF}\left(F^s_4, F^o_4\right). \tag{9}$$

In summary, the spatial detail and semantic information provided by the multimodal data can be effectively exploited through MFF and MSF.

D. Decoder

In this article, we design a building height decoder and a building footprint decoder, which are independently used for building height estimation and building footprint extraction, respectively. The initial idea is that building height information tends to be continuous, while the semantic category is a discrete property. With only one decoder, the building boundary may be negatively affected by the height estimation. Therefore, with a dual-decoder design that does not share weights, it is possible to use the predicted footprint to refine the boundaries of the building height maps, yielding a more accurate 3-D product.

To obtain the pixel-by-pixel results, the top-layer feature of the encoder usually is sampled to full resolution, i.e., the 2-D shape is $H \times W$. For example, FCN [38], PSANet [39], and DeepLabV3 [40] all upsample the feature map of the last layer of the encoder, which is a more common strategy in semantic segmentation models. Moreover, additional processing is added before upsampling to obtain fine node information. For example, DeepLabV3 uses spatial pyramid pooling and convolutional kernels with various dilated rates to extend a more typical representation. Unfortunately, the lack of height response information at lower levels makes it difficult for the regression model to converge. Therefore, we decode the fusion of low-level spatial detail feature with high-level semantic feature by skip connection similar to U-Net. In this way, accurate building height and footprints can be obtained.

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Specifically, each decoder consists of four upsampling decoding blocks and a 1×1 convolution layer. Each upsampling decoding block contains an upsampling layer and a Conv2dReLU structure, where Conv2dReLU is a serial convolution and batch normalization operation. The input to the decoder is $\{F_1', F_2', F_3', G\}$, where F_1', F_2', and F_3' are the multiscale local representations of the fused features of the SAR and optical modalities, and G is a global representation. Initially, G is the input to an upsampling layer to obtain the output G' after adjusting the resolution. Then, G' and the lower layer feature F_3' are connected by skip connection and sent to Conv2dReLU to obtain the cross-layer fusion result G''. They can be formulated as

$$G' = \text{Upsample}(G) \tag{10}$$

$$G'' = \text{Conv2dReLU}\left(F_3' \odot G'\right). \tag{11}$$

The following three upsampling decoding blocks work similarly, taking the result of the previous fusion stage and connecting it to the previous low-level features. By cascading four upsample decoding blocks, the spatial resolution of the feature map is gradually increased to an $H \times W$ consistent with the input image. Finally, we use a 1×1 convolution layer to adjust the number of channels to yield the final pixel-level mask. Note that for the building height decoding branch, we set the category number to 1. For the building footprint decoding branch, the category number is set to 2 (i.e., building and background).

E. Loss Function

We develop a multitask learning loss to optimize building height regression and building footprint segmentation tasks. The mean square error (MSE) loss is a widely used regression loss function that calculates the mean of the squared differences between the predicted and measured values. It has been proved that the MSE loss has a favorable effect on building height estimation [23], [27], so it is used as a regression loss for MF-BHNet. The MSE loss can be defined as

$$\mathcal{L}_{\text{mse}} = \frac{1}{n}\sum_{i=1}^{n}\left(h_i - \hat{h}_i\right)^2 \tag{12}$$

where h_i denotes the reference building height, $\hat{h}_i$ denotes the predicted building height, and n denotes the number of samples.

The cross-entropy loss is currently widely used for semantic segmentation tasks. However, the cross-entropy loss tends to be overconfident and is susceptible to noisy samples. In large-scale building mapping, noisy samples will inevitably exist. Therefore, we adopt label smoothing loss to alleviate the problem of noisy data. Label smoothing loss uses the parameter α to smooth the true label, and the smoothed label is represented as

$$y_i^{\text{LS}} = y_i(1-\alpha) + \alpha/n \tag{13}$$

where y_i denotes the true label, and y_i^{LS} denotes the smoothed true label. The corresponding label smoothing loss is expressed as

$$\mathcal{L}_{\text{lce}} = -\frac{1}{n}\sum_{i=1}^{n} y_i^{\text{LS}} \log\left(\hat{y}_i\right) \tag{14}$$

where $\hat{y}_i$ denotes the predicted label. Meanwhile, the dice loss is introduced to deal with the sample imbalance problem, defined as

$$\mathcal{L}_{\text{dice}} = 1 - \frac{2\left|y_i \cap \hat{y}_i\right|}{|y_i| + \left|\hat{y}_i\right|}. \tag{15}$$

In summary, the full objective can be expressed as

$$\mathcal{L} = \mathcal{L}_{\text{mse}} + \alpha\mathcal{L}_{\text{lce}} + \beta\mathcal{L}_{\text{dice}} \tag{16}$$

where α and β are the hyperparameters that balance the optimization objective of the network.

IV. Experimental Setup

A. Datasets

Following the dataset construction process in [23], a basic building height dataset covering 62 major cities in China is conducted. We collected Sentinel-1 SAR images, Sentinel-2 optical images, and corresponding reference building height and building footprint data. The following is a description of each data product.

1) Sentinel-1/2 Data: The Sentinel-1 satellite provides comprehensive and free global SAR imagery in two polarization bands, VV and VH, with a spatial resolution of 10 m. To ensure the complete coverage of the sample area, we merge data from both orbital directions. We set a minimum backscatter coefficient threshold of -30 dB to exclude speckle noise from the observations. The Sentinel-2 is a high-resolution multispectral imaging satellite consisting of Sentinel-2A and Sentinel-2B satellites, providing 13 bands. For this study, we selected images from band 2 (blue), band 3 (green), band 4 (red), and band 8 (NIR) with a spatial resolution of 10 m. The images underwent radiometric calibration and atmospheric correction to produce the atmospheric bottom reflection data.

2) Reference Building Height Data: The reference building height is derived from publicly available field survey data from Autonavi Map (https://ditu.amap.com/). These data provide the range of buildings and the corresponding number of floors in major Chinese cities and are released in vector form. Especially, the reliability of the data has been verified in the literature [27], with the MSE of 1.19 m. In our experiments, we transformed the number of floors into building height using experience and building codes. The height of each floor in buildings with less than 33 floors is set to 3 m, while buildings with more than 33 floors have a height of 5 m per floor. The data were transformed from vector to raster format with a spatial resolution of 10 m.

3) Reference Building Footprint Data: Additional building footprint data assistance is required to obtain fine-grained boundaries due to partially missing reference building height data. Compared to the dataset used in [23], we extend it to include more detailed building footprint data. The China Building Rooftop Area (CBRA) dataset, which is the first high-resolution (2.5 m) and multiyear (2016–2021) building

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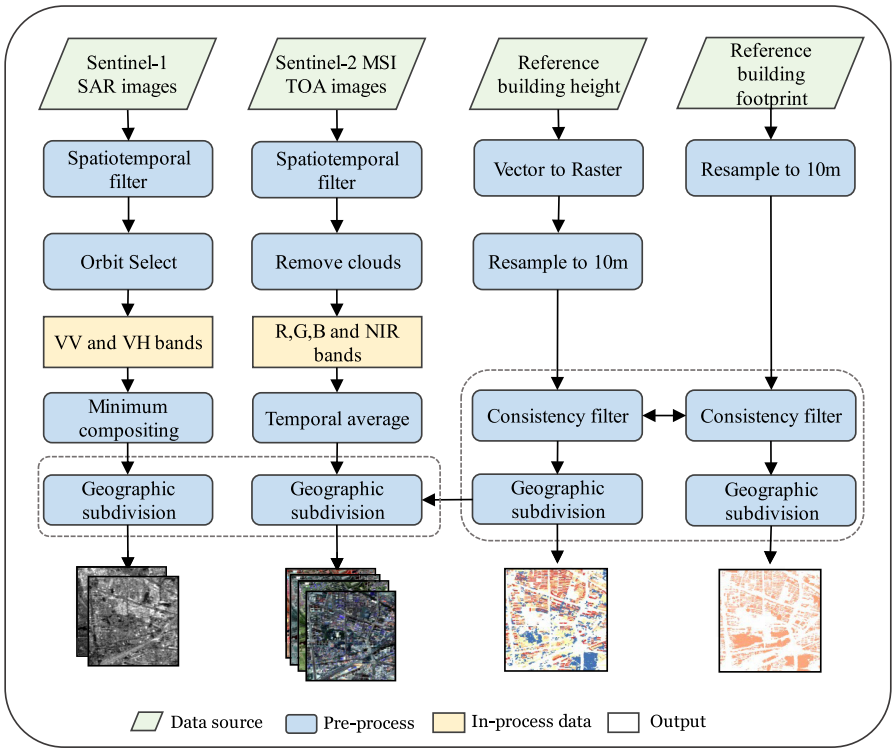


Fig. 6. Flowchart for dataset construction.

roof area dataset in China [41], is selected as the reference footprint data. The accuracy of CBRA has been validated on 250 000 test samples in urban areas. Compared to the WSF2019 dataset used in the literature [23], CBRA has clearer building contours that are not disturbed by roads.

Fig. 6 illustrates the dataset construction process. First, Sentinel-1 and Sentinel-2 images for the year 2020 are obtained from Google Earth Engine (GEE) [42]. Meanwhile, reference building height and footprint data were uploaded to GEE and converted to 10-m resolution raster data. High-quality samples were then retained based on the spatial consistency assessment of the reference building height and footprint data, following the methodology described in [23]. All images were segmented into 256 × 256 pixels tiles. Each sample pair includes Sentinel-1 (VV and VH), Sentinel-2 (R, G, B, and NIR), reference building height data, and reference building footprint data. In addition, the dataset was expanded by including pure image pairs, such as forests and water systems, at a 5:1 ratio, producing a total of 3565 images.

B. Implementation Details

We build and train our network models based on the Pytorch library. The experiments are conducted on an Ubuntu system with an NVIDIA 3090 graphics card with 24 G memory. The AdamW optimizer is used with an initial learning rate of 0.001. The StepLR scheduler is used with a step size of 8 epochs and gamma of 0.95. The epoch and batch size are set to 400 and 32, respectively. The training and validation sets are divided into an 8:2 ratio. During the training process, the hyperparameters α and β, which balance the optimization objective of the model, are both set to 10. Since the building segmentation loss is not in the same order of magnitude as the regression loss, this setting can better facilitate multitask learning.

C. Evaluation Metric

In this section, evaluation metrics are defined to quantify the performance of the proposed method. Mean relative error (Rel), root-mean-square error (RMSE), and threshold accuracy (δ_γ) are defined to evaluate the effectiveness of building height estimation. They are defined as

$$\text{Rel} = \frac{1}{n}\sum_{i=1}^{n}\frac{|h_i - \hat{h}_i|}{h_i} \tag{17}$$

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(h_i - \hat{h}_i)^2} \tag{18}$$

$$\delta_\gamma = \max\left(\frac{h_i}{\hat{h}_i}, \frac{\hat{h}_i}{h_i}\right) < 1.25^\gamma, \quad \gamma \in \{1, 2, 3\} \tag{19}$$

where h_i is the reference building height, $\hat{h}_i$ is the predicted building height, and n is the number of samples.

RMSE is a widely used metric for evaluating regression problems. It provides an overall measure of the deviation between the predicted and actual values. Rel measures the error between the predicted and measured values, while δ_γ measures the degree of proximity between the predicted and reference building height. δ_γ is a measure of the ratio between the predicted building height and the reference building height. It should be noted that lower Rel and RMSE indicate lower error values and higher δ_γ indicates more precise results.

For building footprint segmentation, the mean intersection over union (mIoU) and mean $F1$-score (mF1) metrics are used

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TABLE I
PERFORMANCE OF 12 BUILDING HEIGHT ESTIMATION METHODS. LOWER VALUES OF REL AND RMSE INDICATE BETTER PERFORMANCE, WHILE HIGHER VALUES OF OTHER METRICS INDICATE BETTER PERFORMANCE

Method	Rel↓	RMSE(m)↓	δ_1	δ_2	δ_3	mIoU	mF1
FCN_HE	0.4604	9.9917	0.2728	0.5068	0.6906	0.6697	0.7980
DeepLabV3+_HE	0.4338	8.0328	0.3363	0.6093	0.7321	0.7784	0.8735
U-Net_HE	0.4170	5.1427	0.3595	0.5978	0.7467	0.9032	0.9381
SegNet_HE	0.4321	5.7752	0.2902	0.5210	0.6873	0.8507	0.9185
LinkNet_HE	0.4243	6.5358	0.2749	0.5100	0.6841	0.8632	0.9259
SFNet_HE	0.4302	8.1927	0.3191	0.5728	0.7498	0.7672	0.8663
TransUnet_HE	0.4065	5.0283	0.3657	0.6061	0.7487	0.9083	0.9510
SegFormer_HE	0.4651	9.9326	0.2716	0.4958	0.6914	0.6983	0.8191
Lawin_HE	0.4806	9.2927	0.3046	0.5134	0.7028	0.7309	0.8419
BHE-Net	0.3868	5.3944	0.4117	0.6559	0.7834	0.8746	0.9325
MBHR-Net	0.4320	7.2219	0.3218	0.5835	0.7047	-	-
MF-BHNet(ours)	**0.3251**	**3.6421**	**0.5080**	**0.7135**	**0.8157**	**0.9102**	**0.9527**

to evaluate the performance. The specific formulas are defined as

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{20}$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{21}$$

$$\text{mIoU} = \frac{1}{N_c} \sum_{i=1}^{N_c} \frac{\text{TP}}{\text{TP} + \text{FN} + \text{FP}} \tag{22}$$

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{23}$$

$$\text{m}F1 = \frac{1}{N_c} \sum_{i=1}^{N_c} F1 \tag{24}$$

where TP, FP, and FN denote the number of true positives, false positives, and false negatives, respectively, and N_c denotes the number of categories. For the building height estimation task, mIoU indicates the "shape accuracy" of the height predictions. A higher mIoU means that the building areas are located more accurately, which can lead to more detailed height estimates. It should be noted that we do not mask building heights with predicted building profiles to better reflect the validity of building height estimates.

D. Methods for Comparison

To verify the effectiveness of the proposed MF-BHNet, we compared it with 11 representative methods. Specifically, the BHE-Net [23] and MBHR-Net [21] are the CNN-based building height estimation methods, and both use Sentinel-1/2 data. In particular, BHE-Net is a dual-branch U-Net framework that learns the features of different modalities separately through partially interacting encoders. It adopts two decoders to produce building height and building footprint outputs, respectively. MBHR-Net follows a dual-encoder single-decoder structure. It uses two encoders to extract the SAR image feature and optical image feature, respectively, and then cascades the two modality features. In MBHR-Net, the building height is generated via a single decoder, and the building footprint cannot be directly captured.

Moreover, nine state-of-the-art pixel-level prediction networks were selected for comparative experiments. The current deep learning-based building height estimation models are translated from the field of semantic segmentation. Specifically, FCN [38], U-Net [43], SegNet [44], LinkNet [45], DeepLabV3+ [46], and SFNet [47] using CNN architecture were selected, including SegFormer [48] and Lawin [49], which use transformer framework, and TransUnet [50], which employs a hybrid CNN-Transformer architecture. We redesign the above nine networks with a dual-branch encoder–decoder structure to simultaneously produce building height and building footprint. In the subsequent experiments, we denote them as Net_HE.

V. RESULTS AND DISCUSSION

A. Performance Comparison

1) Comparisons With Other State-of-the-Art Methods: To evaluate the performance of MF-BHNet, we conducted comparative experiments using the constructed dataset. Table I shows the performance of the 12 methods for building height estimation and building footprint segmentation. The best results are highlighted in bold.

Furthermore, compared to BHE-Net and MBHR-Net, which are specialized height estimation networks, the RMSE of MF-BHNet is reduced by 1.752 and 3.579 m, respectively. The current methods rarely consider intramodal discriminative information or fine-grained multimodal information interactions, which limits their performance. In contrast, our method has significant advantages due to the designed multimodal feature semantic fusion module. With progressive fusion from coarse to fine, modality-specific valuable knowledge and modality-complementary information are effectively mined.

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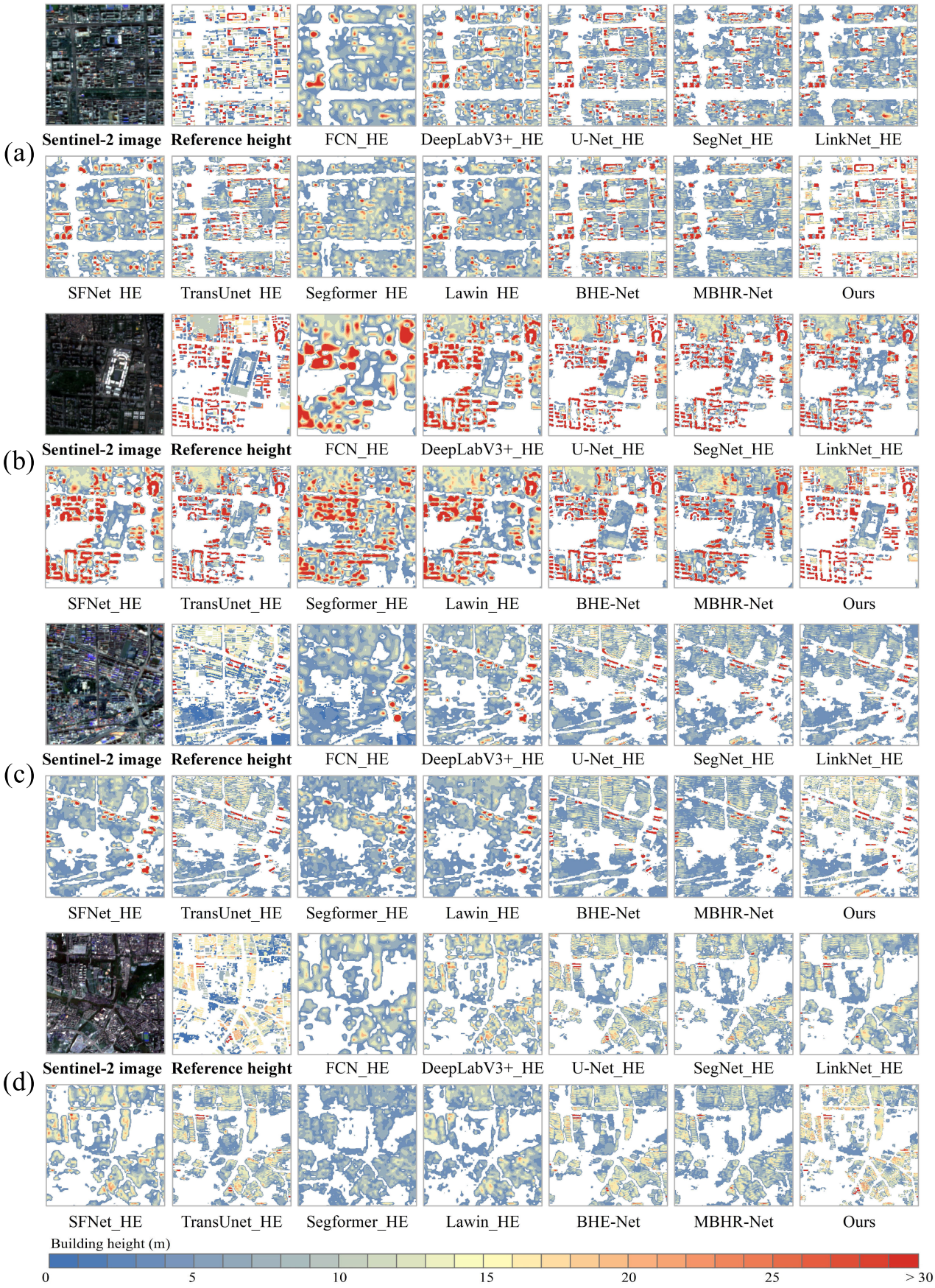


Fig. 7. Building height estimation in China: (a) Beijing, (b) Guangzhou, (c) Baoding, and (d) Shaoxing.

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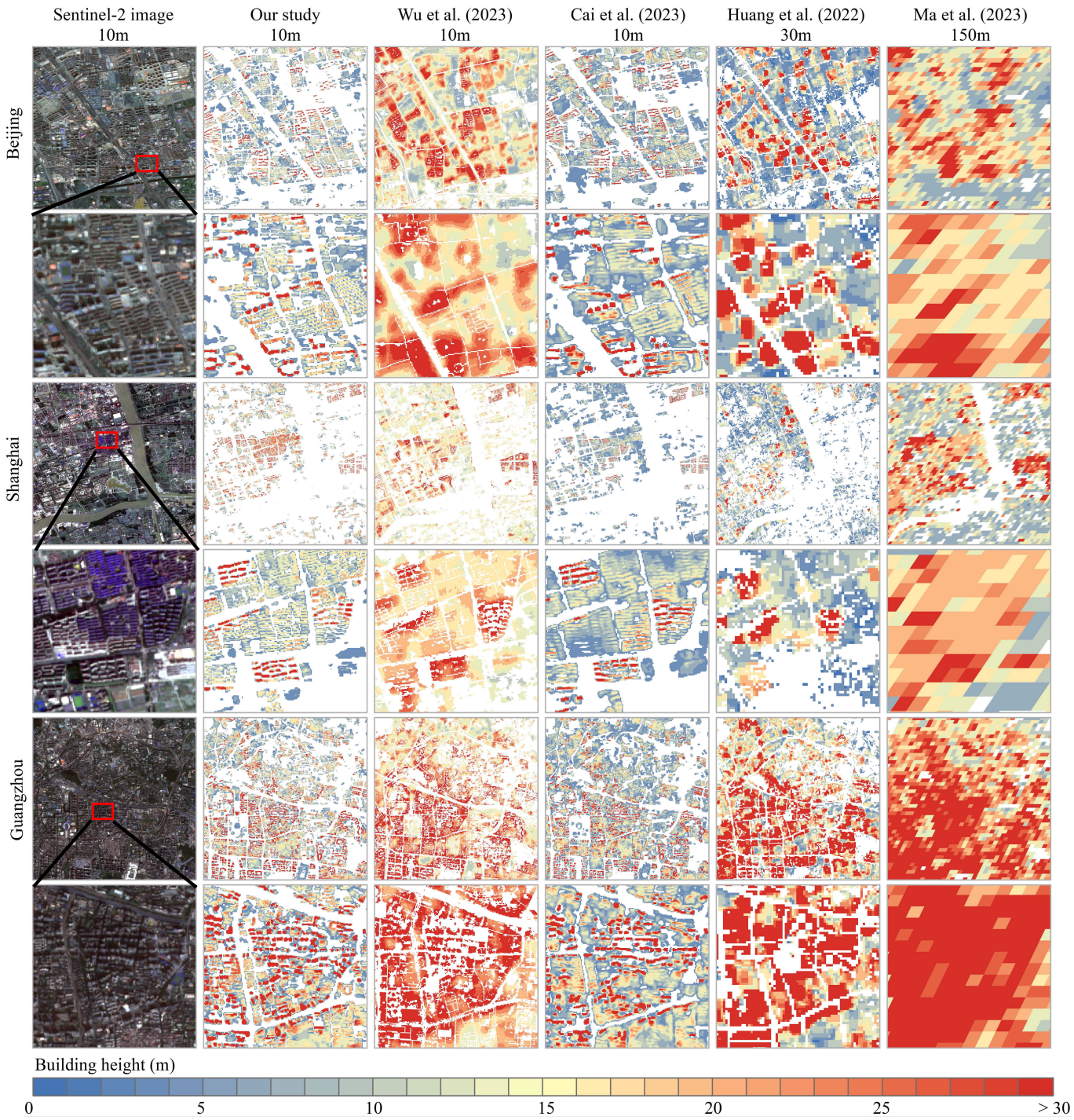


Fig. 8. Intercomparison of building height mapping with Wu et al. [54], Cai et al. [23], Huang et al. [55], and Ma et al. [56].

In addition, we present estimation maps for four distinct urban areas: Beijing, Guangzhou, Baoding, and Shaoxing in China. These areas are chosen due to their varying economic levels and urban height distributions. For example, Beijing and Guangzhou are more developed and have more high-rise buildings, while Baoding and Shaoxing have smaller building floors. Fig. 7 shows the building height mapping results. It can be found that our method is optimal in all scenarios. As shown in Fig. 7(a) and (b), MF-BHNet provides more accurate height and footprint of tall buildings. In contrast, BHE-Net and U-Net_HE, which are better in the comparison methods, have false alarms in the neighboring image pixels of tall buildings. Fig. 7(c) and (d) demonstrates the challenge of distinguishing building boundaries in dense and low-rise areas for the comparison method. Instead, our method provides better spatial detail of building height mapping.

By intramodal critical information mining and multimodal fusion, our method utilizes building boundary information from optical images to refine the distribution of building heights. In summary, the proposed method effectively

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TABLE II
PERFORMANCE OF METHODS WITH DIFFERENT MODULES

Method	Rel↓	RMSE↓	δ_1	δ_2	δ_3	mIoU	mF1
Baseline	0.4070	5.2536	0.3368	0.5663	0.7155	0.8836	0.9238
CME	0.4065	5.0283	0.3657	0.6061	0.7487	0.9083	0.9510
MFF	0.3520	4.1387	0.4637	0.6844	0.7998	0.9072	0.9500
MSF	0.3621	4.6287	0.4537	0.6715	0.7824	0.9012	0.9510
MFF+IME	0.3489	4.0906	0.4692	0.6898	0.8003	0.9070	0.9501
MFF+MSF	0.3285	3.7389	0.4999	0.7108	0.8120	0.9087	0.9504
MFF+MSF+IME+CME (MF-BHNet)	**0.3251**	**3.6421**	**0.5080**	**0.7135**	**0.8157**	**0.9102**	**0.9527**

improves the accuracy of building height estimation and has significant advantages over the compared methods.

2) Benefits of Transformer in MF-BHNet: Encoding–decoding segmentation networks, such as U-Net, which are built on the top of CNN, have become the standard for pixel-level prediction tasks. However, the inherent localization of convolutional operations limits their ability to model explicit global contextual dependencies. Transformer architectures can effectively model global contextual dependencies in images, compensating for the local information modeling capability of convolutional networks. This is crucial for capturing the semantic features of complex urban scenes [51].

The results in Table I demonstrate the significant role of the transformer in our approach. Compared to the CNN-only U-Net_HE method, the RMSE error of MF-BHNet is reduced by 1.5 m. However, simply introducing a self-attention architecture does not always lead to better regression results. For example, the performance of SegFormer_HE and Lawin_HE, which use a purely self-attention design, is lower than that of CNN models such as U-Net_HE and LinkNet_HE. This could be because SegFormer_HE and Lawin_HE model the global context from the lowest level and lack the detailed localization information from low-resolution features, leading to poor building height regression. To address this, MF-BHNet uses vision transformer (ViT) to encode the multimodal feature maps from CNN in chunks, combining the local feature representation and global context dependencies.

Some studies have also explored combining CNN and self-attention for more robust feature learning. For instance, the CNN-Transformer hybrid architecture of TransUnet_HE achieved a suboptimal RMSE of 5.0283 m. However, the metrics such as δ_1 and δ_2 for TransUnet_HE are still unsatisfactory, indicating inaccurate building height predictions with many over- or underestimations. Optimizing the model's architecture alone is limited in accurately predicting building heights. In contrast, our method effectively mines and fuses multimodal information, reducing the RMSE by 1.38 m compared to TransUnet_HE.

Overall, the transformer architecture in MF-BHNet can enhance the model's understanding of the building structure and improve height prediction accuracy. However, it also needs to address the challenges of computational overhead and local information modeling capability. The key direction to improve MF-BHNet is through rational network design and multimodal fusion strategy to fully leverage the advantages of the transformer.

B. Ablation Study

1) Module Validity: To determine the impact of different components in our MF-BHNet, ablation studies are conducted. Specifically, we investigate the impact of four modules: multiscale feature fusion, multimodal semantic fusion, intramodal encoder, and cross-modal encoder, which are denoted as MFF, MSF, IME, and CME. Table II presents the experimental results equipped with different module methods. The baseline represents the raw network ablated of all our improvements.

It can be seen that methods with a single module outperform the baseline. For example, MFF captures richer modal information by fusing multilevel features and reduces RMSE by 21.2% compared to baseline. The MSF reduces the RMSE by 11.9% over baseline, indicating that semantic fusion can facilitate the prediction effect of SAR and optical data.

However, the single module improvement is limited because it only considers the effect of a few factors. When MFF is coupled with MSF, a surprising effect is produced. The RMSE of MFF + MSF is reduced by 28.8% compared to the baseline, while δ_1 is improved by 48.4%. This indicates that the proposed feature fusion and semantic fusion can fully utilize the complementary nature of multimodal data and effectively improve building height estimation. By combining all modules, MF-BHNet achieves the highest δ_1 of 50.8%, indicating that the predicted building height is close to the reference building height. In addition, MF-BHNet also achieves the best building segmentation results, which is an effective multitask learning framework. Experimental results demonstrate that our MF-BHNet can better integrate the advantages of each module to improve the accuracy of building height estimation.

2) Effectiveness of SAR and Optical Data Fusion: It has been found that both optical and SAR imagery can be used for building height estimation, but fusion provides better performance. However, this conclusion is based on the machine learning models, and the deep learning models have stronger feature representation capabilities. Therefore, it is necessary to investigate the performance of deep learning-driven multimodal collaboration. We investigate the performance of band-level fusion, pixel-level fusion, and feature-level fusion methods. For clarity, the following comparison methods are defined as follows.

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TABLE III
PERFORMANCE COMPARISON OF DATA FUSION STRATEGIES FOR BUILDING HEIGHT ESTIMATION

Data		Encoder	Fusion method	Rel↓	RMSE↓	δ_1	δ_2	δ_3
Sentinel-1	Sentinel-2							
√	×	single	-	0.4346	5.4802	0.3507	0.5833	0.7345
×	√	single	-	0.4431	5.9532	0.3164	0.5562	0.7170
√	√	single	Band overlay	0.4034	5.0064	0.3983	0.6290	0.7624
√	√	single	IHS transform	0.4255	5.3192	0.3533	0.5813	0.7258
√	√	single	Laplacian blending	0.4142	5.2225	0.3861	0.6116	0.7504
√	√	double	MSK	0.3965	4.9812	0.4023	0.6359	0.7686
√	√	double	MFF+MSF (Ours)	**0.3285**	**3.7389**	**0.4999**	**0.7108**	**0.8120**

1) *Band-Level Fusion:* Overlaying multiple bands directly is the intuitive and simple way to fuse SAR and optical images. In our experiments, we overlap six different bands into a single six-band input.
2) *Pixel-Level Fusion:* Hue-intensity-saturation (HIS) transform and Laplacian blending are selected as pixel-level fusion methods. In particular, the HIS transform modifies the histogram of a high-resolution SAR image according to the statistical parameters of the intensity components and converts the modified components into an optical image to obtain a fused image [52]. The Laplacian blending selects fusion windows of varying sizes for different frequency components [53].
3) *Feature-Level Fusion:* The MSK [23] is a feature-level fusion method for mapping building height, which refines the cascade of SAR and optical features. The MSF and MFF modules we designed also belong to feature-level fusion.

In addition, we present the regression performance using only SAR or optical imagery. To apply different fusion methods for building height estimation, we design single-encoder- and dual-encoder-driven networks for the single-input and dual-input cases, respectively. Table III shows the performance of different fusion methods in the building height estimation task. MFF + MSF stands for the method that combines the proposed MFF and MSF modules.

The results show that either SAR or optical imagery can be used separately to estimate building height. However, the performance of the single modality is limited by the absence of essential supporting information. Simply overlaying six bands improved performance, improving the RMSE by 8.65% compared to SAR data alone and by 15.9% compared to optical data alone. The combination of SAR and optical data can provide richer information and better results. However, the coarse fusion scheme does not make effective use of the important information within the modality and may be interfered with by redundant information. The IHS transform and Laplacian blending fuse pixel-level information of SAR and optical images to some extent. However, pixel-level fusion does not achieve better results than band cascading since the visual representation of SAR and optical data is quite different.

TABLE IV
PERFORMANCE OF BUILDING HEIGHT ESTIMATION WITH DIFFERENT BACKBONES

Model	Baseline	MFF	MSF	RMSE↓
LinkNet_HE	✓			6.5358
	✓	✓		5.7566
	✓		✓	5.8245
	✓	✓	✓	5.3915
DeepLabV3+_HE	✓			8.0328
	✓	✓		7.0231
	✓		✓	7.2514
	✓	✓	✓	6.4299

As for feature-level fusion, the MSK refines the features of SAR and optical cascades, which performs better than other methods compared. However, it concatenates multimodal features directly, neglecting the careful combination of exclusive and semantic information within modalities. In contrast, MFF + MSF improves building height estimation accuracy through feature fusion and semantic fusion. Compared to the suboptimal method, our fusion method improves δ_1 by 21.2%, while the MSE is reduced by 24.9%. Overall, our method takes advantage of different data modalities to better collaborate in the building height mapping task.

3) Backbone Adaptability: To validate the applicability of the proposed multimodal feature-semantic fusion module, we replace the original encoder–decoder network with LinkNet and DeepLabV3+. LinkNet combines encoder–decoder connections with residual blocks, improving the efficiency of information flowthrough the network. In contrast, DeepLabV3+ employs dilated convolution to expand the receptive field. In terms of implementation, similar to MF-BHNet, different fusion modules are embedded within the encoder. Table III presents the performance of building height estimation using the LinkNet_HE and DeepLabV3+_HE networks.

As shown in Table IV, the performance of the LinkNet_HE significantly improves when incorporating both the MFF and MSF modules. The RMSE values decrease by 0.7792 and 0.7133 m, respectively, compared to the baseline. For the DeepLabV3+_HE, the MFF and MSF modules reduce the

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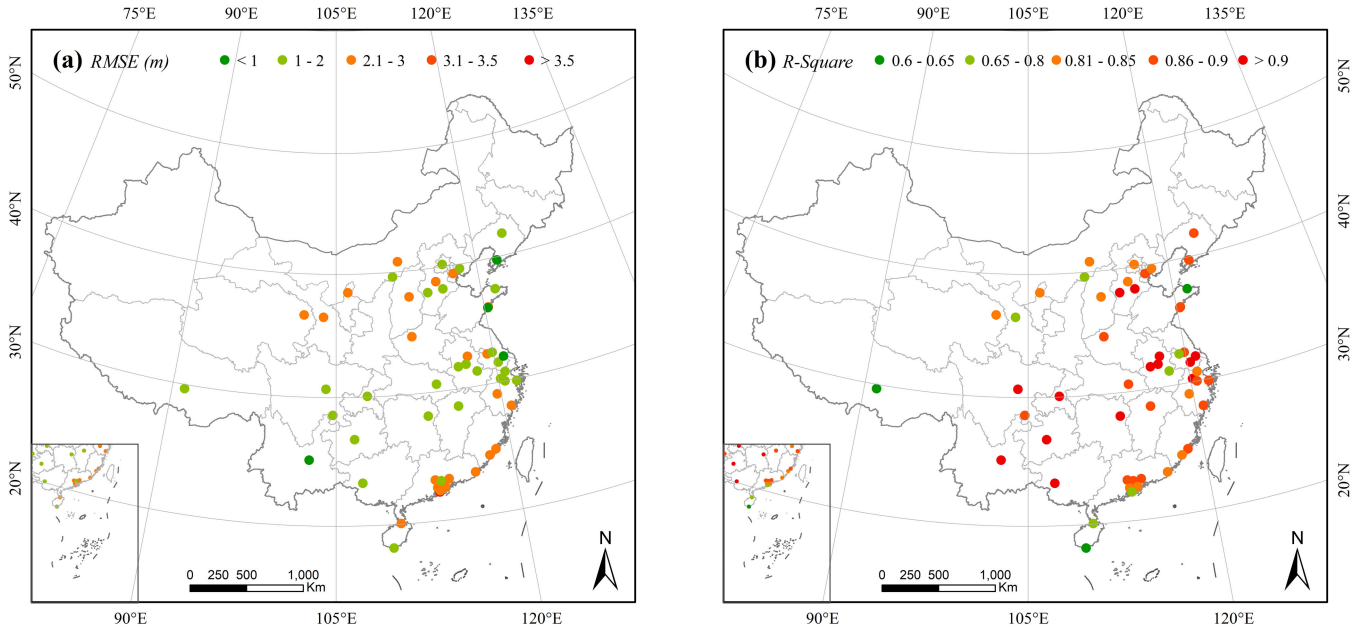


Fig. 9. Validation of building height of 55 cities in China. (a) RMSE distribution. (b) R-square distribution.

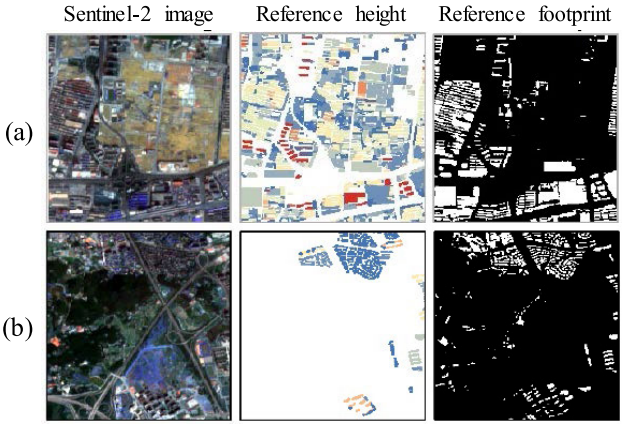


Fig. 10. Noisy samples. (a) Reference height data mismatch. (b) Reference height data partially missing.

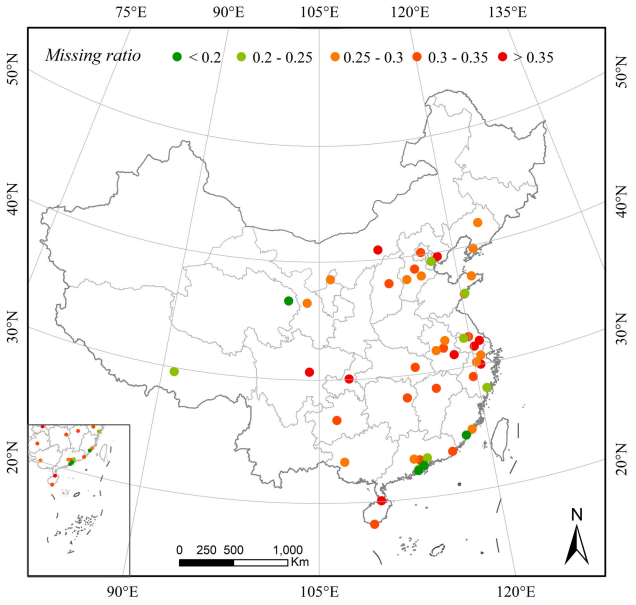


Fig. 11. Missing ratio of reference building height data to building footprint data.

estimation errors by 1.0097 and 0.7814 m, respectively. Moreover, the combination of MFF and MSF modules yields even greater enhancements in model performance. When compared to the baseline, the MFF + MSF integration reduces the RMSE by 1.1443 and 1.6029 m for the two backbone models, respectively. This demonstrates that the proposed fusion modules can be effectively adapted to various backbone networks while maintaining strong performance.

C. Potential of MF-BHNet for Building Height Estimation

1) Intercomparison With Recent Building Height Products: To assess the potential of MF-BHNet for large-scale mapping, we mapped building heights in three test areas selected from the Chinese cities of Beijing, Shanghai, and Guangzhou. We compared the maps produced with four publicly available building height products. Specifically, the compared products include the Chinese building height map (CNBH) by Wu et al. [54] and BHE-Map by Cai et al. [23] at 10-m resolution, the 30-m product by Huang et al. [55], and the 150-m GBH-2020 by Ma et al. [56]. Fig. 8 presents the building height maps for three urban areas in China: Beijing, Shanghai, and Guangzhou.

The results indicate that our method produces more detailed building height maps than the compared products. CNBH relies on shadow features, which cannot effectively distinguish the boundaries of dense buildings. Moreover, the traditional machine learning methods have limited estimation capability. The deep learning-driven BHE-Map is consistent with our building height trends. However, the building boundaries of

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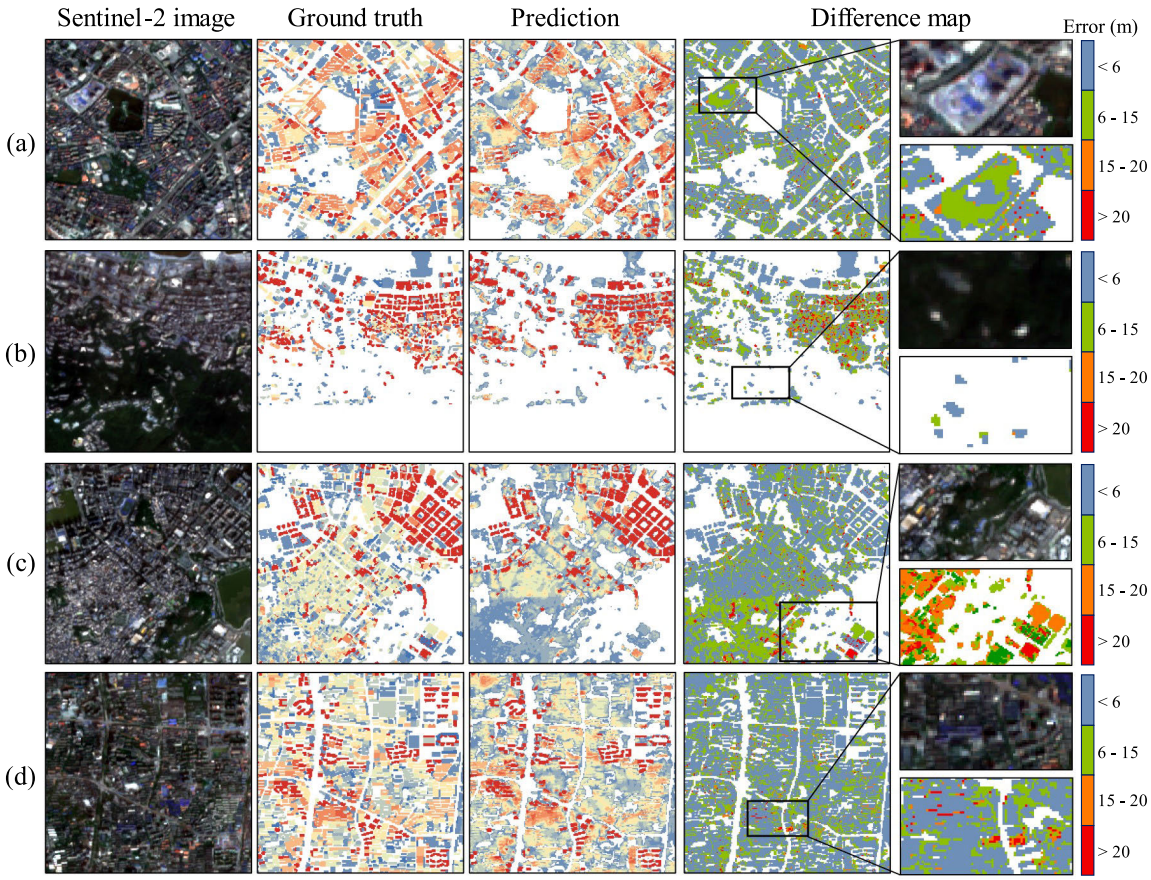


Fig. 12. Difference between predicted and ground truth. Four regions from (a) Wuhan, (b) Hong Kong, (c) Macau, and (d) Changsha.

BHE-Map are more blurred and there are some false positives between neighboring buildings. The 30-m product by Huang et al. [55] and the 150-m GBH-2020 tend to cluster image pixel heights and cannot distinguish between buildings.

In summary, our method demonstrates superior mapping results compared to published building height products. As an end-to-end mapping method, it provides a novel and more efficient solution for rapidly mapping building height at national and even planetary scales.

2) Generalizability in Different Cities of China: We visualize the relationship between reference building heights and estimated building heights for 55 cities in China to reveal the generality of MF-BHNet to different geographical areas and urban layouts. These cities have relatively complete reference data of high quality. Fig. 9 shows the RMSE and R-square (R^2) for different cities.

Overall, most of the cities exhibit high goodness-of-fit and low prediction errors, indicating the generalizability of our method across different cities. Specifically, 15 cities have RMSE values below 1 m, demonstrating very high prediction accuracies of the model in these cities. These cities include megacities such as Beijing, Shanghai, Guangzhou, Shenzhen, and other large cities such as Hangzhou, Tianjin, Wuhan, and Xi'an. This suggests that the MF-BHNet model performs exceptionally well in high-density urban building areas. Furthermore, the R^2 values of 20 cities exceed 0.9, meaning that the model is able to explain more than 90% of the building height variability in these cities.

In contrast, Macau has the largest RMSE value of 3.906, while Lhasa, Yantai, and Sanya have the lowest R^2 values of less than 0.65, indicating a relatively weaker fitting effect of the model in these locations. Macau, Sanya, and Yantai are situated in coastal areas with complex and variable topography. The influence of island/peninsular geography on building forms may affect the estimation accuracy of the model. The buildings in Lhasa are located in the mountainous highland area, with a complex topographic environment. Therefore, it may be necessary to explore the integration of additional geographic data (e.g., topographic information) to enhance the model's ability to predict building heights in complex geographic environments.

D. Limitation and Future Study

1) Analysis of Limited Reference Data: Similar to other published studies, the building height estimates in this work rely on reference data. However, the reference building height data and the reference footprint data came from different sources in this study. Our observations indicate that the reference footprint data have a high degree of completeness, while the reference height data have significant deficiencies, as shown in Fig. 10. To explore the inconsistency in the reference height data, we use the reference building footprint data as the ground truth for the distribution of buildings.

Fig. 11 presents the percentage of missing data in the reference building height data for different cities. The results show

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that most cities have inconsistency values between 0.25 and 0.35. Some cities, such as Eerduosi, Loyang, and Ningbo, have more than 35% of the reference height data missing, meaning that over one-third of the building height information is unavailable for these cities. To avoid this inconsistency from misleading the model, we applied a masking operation during model training to ignore the effect of the inconsistent image pixels on the loss calculation. For future work, methods such as transfer learning or semisupervised learning can be explored to supplement the training data for cities with severe missing height data.

2) Failure Cases: The aforementioned experiments demonstrate that our method achieves the most competitive performance. Inevitably, our method also produces false positives in a few regions. Thus, we conduct a detailed error analysis to identify where the model underperforms and potential areas for improvement. Specifically, we select a few regions, where the RMSE is greater than 5 m, and generate plots of the absolute error between the predicted results and the ground truth. Fig. 12 shows the Sentinel-2 imagery, predictions, ground truth, and difference maps for the different regions.

The predicted results are generally in good agreement with the true values in the overall trend. However, there are also issues of omissions, false positives, and underestimations. For example, in Fig. 12(b), the buildings on the small islands around the land are omitted. This is due to the low spatial resolution of the Sentinel-2 imagery, which makes it difficult for small buildings on isolated islets to be clearly recognized. Meanwhile, there is a problem of underestimation of some buildings, as shown in the lower left of Fig. 12(c). We found that the high and undulating terrain in this area may cause the SAR sensor to not fully receive the backscattered signals from the buildings, which affects the strength of the backscattering. Furthermore, interference from extraneous background features can also affect the building extraction.

In addition, inaccuracies in the dataset can also cause problems for the model. For example, in the selected zoomed-in view image of Fig. 12(a), the reference building height data and the image data are not consistent. The reference building heights were not collected at a time cross section; hence, there is data noise. This may cause the model to move in the wrong optimization direction.

To address the above issues, image super-resolution, multisource data integration, and noise-robust learning are promising future research directions. Image super-resolution methods can be used to improve the discriminative ability of building structures. The integration of multisource geographic data, such as land cover and terrain information, can be explored to improve the prediction accuracy of the model. In addition, noise-robust learning can also be considered as a potential method that contributes to the optimization of the feedback correction model.

VI. Conclusion

In this article, we present a novel hybrid multimodal fusion network for mapping accurate building height with Sentinel-1 SAR and Sentinel-2 optical imagery. First, we design a hybrid multimodal encoder to extract intramodal valuable information and model intermodal correlation. In this way, better intermodal feature fusion is facilitated by perceiving modal information more comprehensively from local to global. Furthermore, a coarse-fine progressive multimodal fusion method is proposed to bridge the heterogeneity problem of SAR images and optical images. Through the coarse-to-fine multiscale feature fusion and MSF, the complementary information provided by SAR and optical data can be effectively integrated and fused. Besides, a building height dataset is constructed by introducing new building footprint data. The experimental results show that our method outperforms the suboptimal method by 27.6% in RMSE, compared with 11 state-of-the-art methods. Moreover, comparing the four publicly available building height products, the mapping of the proposed method has significant advantages in spatial detail and accuracy.

Our method has only been tested in selected cities in China. In the future, our work will be extended to the national scale to perform large-scale building height mapping. Furthermore, satellite remote sensing imagery is currently considered, while crowdsourced geographic information data are useful for building height estimation and validation. We will introduce crowdsourced data to compensate for the mapping limitations of remotely sensed imagery.

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